# Armadale Kalamunda Group

## Oxygen Therapy Management in Adults Guideline

## 1. Purpose

The purpose of this guideline is to establish minimum practice standards for the care and management in adults for oxygen therapy.

## 2. Applicability

This guideline applies to medical, nursing, and allied staff providing care for patients at Armadale Kalamunda Group.

## 3. Guidelines

### 3.1 Safety precautions

**No patient should be denied oxygen therapy in an emergency.**

Patients commenced on acute oxygen therapy should be assessed promptly, carefully and regularly.

Oxygen supports combustion, and there is a risk of fire or explosion, especially if patients or carers should smoke or light matches near the oxygen source. No naked flames or friction toys must be within 2 meters of an oxygen cylinder or delivery equipment including wall outlets, mask and tubing in use

Oxygen cylinders must be stored in a secure, cool, dry place away from heat and direct sunlight.

Oxygen therapy can have a drying effect on oral mucosa. Regularly assess oral mucosa, offer fluids and perform regular mouth care on the patient.

### 3.2 Indications:

| Indication | Examples | Initial O2 Therapy | Target Oxygen Saturation |
|---|---|---|---|
| Critical Illness | Major Trauma<br>Shock<br>CO poisoning | Reservoir (non rebreather mask 15 L/min) | 94% -98% |
| Suspected Type I (hypoxemic) respiratory failure | Pulmonary embolism<br>Pulmonary oedema<br>Pneumonia<br>Diffuse lung disease | Nasal prongs 2-4 L/min or<br>Hudson/simple face mask 5-10 L/min | 94-98% |

| Indication | Examples | Initial O2 Therapy | Target Oxygen Saturation |
|---|---|---|---|
| | Severe asthma | Consider use of Non-rebreather mask<br>CPAP | |
| Suspected Type II (hypercapnoeic) respiratory failure | Severe COPD<br>Severe bronchiectasis<br>Morbid obesity<br>Sleep apnoea<br>Neuromuscular disease<br>Kyphoscoliosis or other chest wall deformity | Nasal prongs<br>High Flow nasal prongs (HFNP)<br>Bilevel positive airway pressure (BIPAP) | 88-92% |
| Conditions where $O_2$ therapy is not required (unless the patient is hypoxaemic) | Coronary artery disease<br>Stroke<br>Pregnancy and obstetric emergencies<br>Drug overdoses | Nil | 94-98% |

### 3.3 Assessment

At a minimum the following physiological initial and regular observations must include:

- Respiratory rate
- Oxygen saturation
- Heart rate
- Blood pressure
- Temperature
- Level of consciousness (Alert, Voice, Pain, Unresponsiveness, AVPU)

Oxygen saturation on room air should be monitored for at least 5 minutes after discontinuing oxygen therapy and should be rechecked in an hour.

### 3.4 Patient management during oxygen therapy

- Ensure tubing connections are secure and oxygen supply is connected and intact
- If a face mask is sitting comfortably over the nose and under the chin
- If nasal cannula are used ensure that a prong is inserted into each nostril and assess mucosal dryness of nasal passages at least once per shift
- Assess skin for evidence of pressure areas behind the ears and on the bridge of the nose area as applicable. Document adverse findings in health records
- During eating and drinking it may be necessary to temporarily administer oxygen via nasal cannula if necessary
- Assess hydration status especially dryness of mucous membranes and consider humidified oxygen. Promote adequate fluid intake whilst considering fluid restrictions

- Perform oral hygiene 4- 6 hourly and prn
- Replace respiratory device as required when soiled or contaminated due to secretions or inadequate storage
- Dispose of mask device in the general waste when patient is discharged/ deceased, or mask is replaced

## 4. Equipment

- Choice of equipment must be appropriate for the age and/or size of the patient
- Equipment must be service in accordance with manufacturer’s recommendations to ensure reliability and accuracy
- Oxygen delivery device
- Monitoring equipment such as oximeter
- Wall oxygen outlet or portable oxygen cylinder with nipple
- Regular oxygen flow meter (0- 15 L/min)
- Low flow oxygen flow meter (0- 2.5 L/min)
- High Flow Oxygen flow meter (0-7L/min)
- Humidification device as necessary

When a low flow meter is in use, a regular flow meter is to remain by the oxygen supply at all times in case of medical emergency. **High Flow oxygen flow meters (0-70L) should only be used in high acuity areas such as ICU and Emergency department.**

## 5. Oxygen prescription

Oxygen therapy is to be prescribed once the patient is stable, except:

- Areas of critical care such as intensive care, coronary care, high dependency and trauma high acuity units
- Critically unwell patients in the emergency department
- Operating theatres
- Immediate post procedure recovery- up to 2 hours following discharge from theatre recovery to the ward with time commencing at the point of discharge from recovery (not the end of the procedure). Oxygen should be weaned off as clinically appropriate. If the patient still requires oxygen after 2 hours they should be reviewed by a medical officer and oxygen prescribed if clinically appropriate. Use nasal prongs if the patient does not tolerate a Hudson ™/ face mask
- Those on pain service therapies (e.g. intravenous opioid infusions, epidural) with oxygen administration as per general management prescription on the relevant infusion chart. If target oxygen saturation range is not maintained escalate care to medical review
- Physiotherapy and respiratory/ sleep laboratories
- Procedures performed outside of main theatre or that require sedation only. **Except** if it is expected that a patient will require ongoing oxygen therapy post procedure

The prescription should be reviewed daily and should define:

- Indication
- Target oxygen saturation

- Delivery device
- Range for oxygen flow or percent of inspired oxygen
- When oxygen is to be applied
- Commencement date
- Signed and dated with prescribers' name printed legibly

A sticker should be placed on the patient's medication chart which will either be the oxygen prescription or will indicate that there is a separate oxygen prescription chart in place.

### 5.1 Oxygen delivery devices

| Delivery Device | Abbreviation | Flow (L/min) | Approx $FiO_2$ (%) |
|---|---|---|---|
| Simple nasal cannula | NP | 0.5- 4 | 22- 33 |
| Low flow nasal prongs/ cannula (use with low flow meter) | LFNP | 0.25- 2 | 22- 28 |
| High flow (humidified) nasal prongs/ cannula – AIRVO2 | NHF | 6- 40 | 21- 60 |
| High flow (humidified) nasal prongs/cannula - HAMILTON | NHF | 6-40 | 21-100 |
| Simple (Hudson) mask | HM | 5- 10 | 40- 60 |
| Reservoir (non-rebreathing) mask | NRM | 10- 15 | 60- 90 |
| Positive airway pressure mask i.e. CPAP* or NIV-ST (BIPAP) | NIV | 0.5- 60 | 22- 60 |
| Swedish nose via tracheostomy | SN | 0.5- 4 | 22- 50 |

### 5.2 Dose of oxygen therapy

- Prescribe at a dose to achieve normal or near normal saturation of 94 - 98% for acutely ill patients
- Titrate to the lowest concentration that meets oxygenation goals:
- 88 - 92% for patients with or at risk of hypercapnoeic respiratory failure
- 94 - 98% for all other patients
- Should be reduced if the patient is clinically stable and the oxygen saturation is above the target range or has been at the upper end of the target range for at least 4 hours. This may require a change to the oxygen delivery device
- Must be ceased when the patient is able to maintain oxygen saturation in the target range when breathing room air. Oxygen saturation on room air should be monitored for at least 5 minutes after discontinuing oxygen therapy and should be rechecked in an hour. Recommence if saturation level falls below the target range

### 5.3 Nasal cannula (simple)

- Nasal cannula can be used to deliver low and medium dose oxygen concentrations. Delivers approximate $FiO_2$ between 0.24-0.33.
- **Not indicated** if the patients clinical state is **unstable**.
- Unsuitable for concentrations greater than 35%.
- Mouth breathing does not appear to reduce the efficacy of nasal cannula.

Generally preferred by patients over face masks and have the advantages of improved comfort, less claustrophobia, ability to eat and speak freely, less easily dislodged, less inspiratory resistance and no risk of (carbon dioxide) $CO_2$ rebreathing.

Disadvantages:

- Flow above 4L/min tends to cause nasal dryness and discomfort if maintained for several hours and may not be effective in patients with severe nasal obstruction 1
- Should not be used in patients who have nasal trauma, nasal injury including epistaxis and nasal blockages.

| | Oxygen Flow per min | Approximate $FiO_2$ |
|---|---|---|
|  | 1 litre<br>2 litres<br>3 litres<br>4 litres | 0.24<br>0.28<br>0.32<br>0.36 |

### 5.4 Nasal cannula flow rate and approximate $FiO_2$

Application of nasal cannula

- Insert nasal cannula into the opening of nares
- Secure the tubing behind the ears ensuring that it is not too tight to cause trauma
- Gently secure under the chin using sliding adjustment piece
- Turn oxygen flow meter on to the prescribed flow rate
- Titrate the flow rate according to saturation level in relation to the medical prescription. Select minimum flow rate to achieve target saturation level

### 5.5 Simple face masks (Hudson Mask)

- A Hudson™ face mask is a simple facemask with open side ports that allows room air to enter the mask and dilute the oxygen as well as allowing exhaled carbon dioxide to leave the containment space.
- Simple face masks deliver oxygen concentrations between 0.40 - 0.60 $FiO_2$.

- Flow must be at least 5 litres per minute to prevent the patient re-breathing $CO_2$. The mask is suitable for patients with hypoxaemic (Type I) respiratory failure but not for patients with hypercapnoeic (Type II) respiratory failure [1].

| | Oxygen Flow per min | Approximate $FiO_2$ |
|---|---|---|
|  | 5- 6 litres<br>6- 7 litres<br>7-10 litres | 0.40<br>0.50<br>0.5-0.6 |

Hudson ™ mask flow rate and approximate $FiO_2$

### 5.6 Application of simple mask

- Apply simple mask to patient
- Set flow rate to minimum of 5L/min [1]
- Do not occlude side ports
- Titrate the oxygen flow rate to the minimum prescribed flow rate, to achieve the target saturation

### 5.7 Non rebreather mask and bag

- Non rebreather (reservoir) masks are most suitable in an emergency where $CO_2$ retention is less relevant [1,2].
- They achieve close to 100% oxygen at 10-15 litres per minute by minimising room air entrapment. They deliver the highest concentration of oxygen without a sealed system (i.e. CPAP, NIV).
- They have multiple one way valves in the side ports and a reservoir bag attached.
- The valves:
- Prevent air being drawn into the mask
- Enables $CO_2$ to leave the mask thus preventing risk of rebreathing

#### 5.7.1 The reservoir bag:

- Fills with a greater concentration of oxygen available for the patient to inspire
- Has a one way valve preventing exhaled air from being rebreathed

#### 5.7.2 Application of non-rebreather mask and bag

- Apply facemask to patient
- Set flow rate to a minimum of 10L/min
- Ensure that the reservoir is inflated at the end of inspiration at all times. If not, check tubing connections and oxygen. If still not inflated, increase flow rate and escalate care. Deflation suggests a leak or inadequate oxygen flow and may be a sign of deterioration [1]

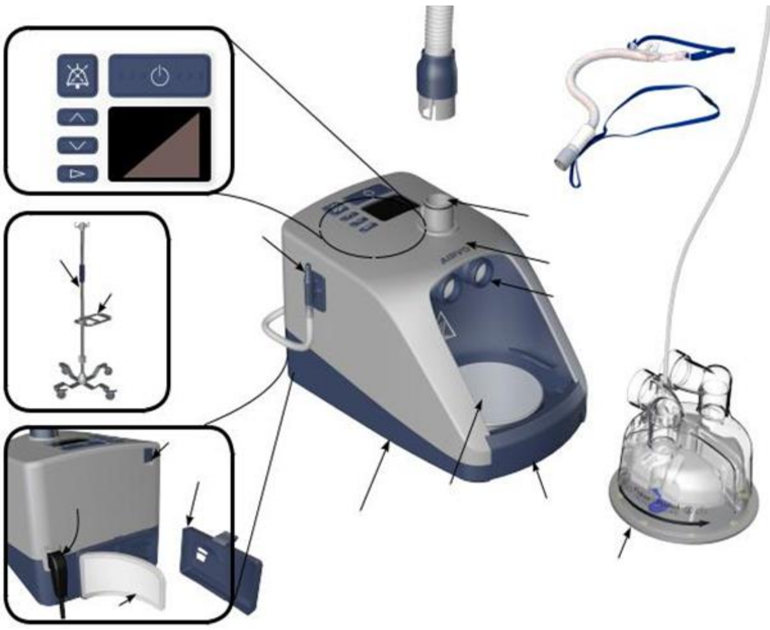
- Titrate the oxygen flow to the minimum prescribed flow rate, to achieve the target saturation. Escalate care if patients condition deteriorates or patient is requiring increasing oxygen requirements to maintain prescribed saturation levels.

### 5.8 Humidification

- Humidification may not be required in patients with intact upper airways using low flow oxygen therapy or short-term use of high flow oxygen.
- Humidified oxygen is required when the upper airway is bypassed by tracheostomy or another artificial airway. A passive or active humidification system (HME) should be used. A heated pass over humidifier should be used for:
- Patients using high flow nasal cannula
- High flow oxygen via face masks used for more than 24 hours
- Patients who report upper airway discomfort due to dryness when using oxygen therapy
- Recent hemoptysis
- Sputum retention

### 5.9 High flow nasal prongs/ cannula (humidified)

- High Flow Nasal Prongs (HFNP) can be used to deliver a user set Fixed Fio2 at high flow rates.
- The AIRVO™2 system uses a humidifier with an integrated flow generator that delivers high flow, warmed and humidifier respiratory gases to assist spontaneously breathing patients. The AIRVO™2 contains an oxygen analyser to help you determine the oxygen fraction you are delivering to your patient. Adjust the level of oxygen from the oxygen source, until the desired oxygen fraction is displayed oxygen fraction is displayed on the screen. It may take the reading several minutes to settle.
- The Airvo™2 can be set up with High flow oxygen meters (up to 70L) but this is only for use in high acuity areas such as ICU or Emergency department.
- The system can deliver up to 60L Flow and humidified oxygen which can deliver oxygen up to 55% in general wards and up to 100% in critical areas.



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### 5.9.1 Indications:

- Hypoxia
- Increased effort of breathing
- To achieve optimal humidification

### 5.9.2 Contraindications:

- Untreated pneumothorax
- Penetrating lung injury

Humidification prevents the nasal mucosa from drying.

### 5.9.3 Key points

- Initiating and maintaining treatment depends on underlying pathology
- Nurse to commence therapy using settings of $FiO_2$ flow and temperature
- Nurse to titrate treatments according to arterial blood gas, venous blood and oximetry results in consultation with medical officer
- Flow rates and FiO2 can be adjusted as clinically indicated but if greater than 40% then escalate care to medical officer and to the CNS. If Fio2 goes above 50% patient must have ICU input in consideration with patients' goals of care
- Patients who are at risk for aspiration and poor swallow who require HFNP oxygenation should have the flow reduced to 30-35L /minute while eating and drinking

| ADDS SCORE | FIO2 on HFNP |
|---|---|
| • 3 | • >40% |
| • 2 | • 34-40% |
| • 1 | • 26-33% |
| • 0 | • 21-25% |
| If Fio2 requirement >40% consider medical and CNS review. Fio2>50% needs ICU review -consider Goals of Care | |

- The HFNP provides an alternative to conventional low flow oxygen therapy. HFNP reduces the inspiratory resistance and prolongs the expiratory phase which promotes slow and deep breathing. It also helps in promoting alveolar ventilation and tidal volumes.
- In patients who do not tolerate the High flow nasal cannula humidified oxygen therapy can be given by attaching the aerosol face mask to a fisher and Paykel interface adaptor and then connecting to a AIRVO™2 machine
- The decision and regimen for weaning a patient off high-flow nasal cannula (HFNC) therapy is based on clinical judgment, which takes into consideration the patient's underlying pathology, respiratory function assessment, trends in arterial or venous blood gases (ABG/VBG), and vital signs. This process should be approached with a multidisciplinary team. Respiratory function assessment involves evaluating parameters such as respiratory rate, effort, oxygen saturation, and lung function tests, if available. This assessment helps determine the patient's ability to maintain adequate oxygenation and ventilation without HFNC support. Vital signs monitoring, including heart rate, blood pressure, and temperature, offers additional information about the patient's overall condition and response to weaning.

## 5.10 Non-Invasive Ventilation – Continuous Positive Airway Pressure (CPAP) and Spontaneous Timed breath (NIV-ST)

- CPAP provides continuous positive pressure throughout the respiratory cycle by recruiting collapsed alveoli, helping to increase functional residual capacity (FRC) and optimize oxygenation. CPAP is effectively used to treat patients with hypoxic respiratory failure without hypercapnia (Type 1 respiratory failure), caused by acute pulmonary oedema and atelectasis, and is also employed in cases of obstructive sleep apnoea (OSA).
- The NIV-ST mode utilizes bilevel positive airway pressure for inspiration and expiration, providing inspiratory positive airway pressure (IPAP) and expiratory positive airway pressure (EPAP) respectively. EPAP is synonymous with positive end-expiratory pressure (PEEP). IPAP aids in ventilation while EPAP assists in oxygenation. The numerical difference between IPAP and EPAP is called pressure support (PS). This mode is utilized for patients with hypercapnic respiratory failure (Type 2 respiratory failure) and aids in increasing minute ventilation, tidal volume, and reducing the work of breathing. **The Non-invasive ventilation should only be used in a critical care environment (ED, ICU, HDU) and requires one on one nursing care.**

### 5.10.1 Oxygen Therapy via a tracheostomy

- All oxygen delivered via a tracheostomy must be humidified to a 37 degree as it bypasses the natural humidification of the upper airway.
- A Swedish nose is a single use heat and moisture exchange (HME) device which allows humidification of all inspired gases for the administration of oxygen via tracheostomy. The central port allows suctioning and sampling

Tracheolife ™ (swedish Nose) humidifier with oxygen vent

#### 5.10.2 Patient Oxygen Management

- Ensure tubing connections are secure with oxygen supply are connected and are intact. Ensure there are no leaks in the tubing and the tubing is connected to a working oxygen supply.
- If a face mask is used ensure a good seal at the bridge of the nose and below the chin
- Always assess the patients need for oxygen therapy while eating and drinking. Assess the hydration status especially drying of the oral mucosa membranes. Patients on continuous oxygen therapy can cause dryness of the oral and nasal mucosa.
- Promote adequate fluid intake -Consider fluid restrictions. Frequent oral fluids may alleviate symptoms of a dry mouth.
- The continuous need for oxygen should be reviewed daily by the nursing team, physiotherapy team and the medical team. A daily check on the skin integrity and assess for pressure injury such as evidence of redness and altered skin integrity.

#### Oxygen Weaning

- Oxygen therapy should be reduced and discontinued in stable patients with satisfactory oxygen saturations
- Dose should be reduced if patient is clinically stable and the oxygen saturation is above the target range or has been at the upper end of the target range for at least 4 hours. This may require a change to the oxygen delivery device
- Oxygen therapy should be ceased when the patient is able to maintain oxygen saturation in the target range when breathing room air. Oxygen saturation on room air should be monitored for at least 5 minutes after discontinuing oxygen and should be rechecked at 1 hour or as clinically indicated. Recommence therapy if the oxygen saturation falls and remains below the target range for 5 minutes
- All the weaning attempts should be notified to the medical officer and if possible, should be attempted while patient is fully awake. Nighttime factors can affect weaning e.g.: sleep apnea, position of the patient and sleep patterns.
- Oxygen should be weaned based on the individual clinical need of the patient and based on the target saturations as decided and prescribed by the medical staff.
- If oxygen therapy is unable to be weaned successfully liaise with the medical officer
- The medical officer should cease the oxygen prescription chart if the patient does not fit the criteria for supplemental oxygenation and when the clinical decision is made to discontinue the therapy.
- Document all attempts at weaning in the Digital Medical record (DMR)

## 6. Special Considerations:

### 6.1 Pregnancy

All women with hypoxaemia who are more than 20 weeks pregnant should be managed with left lateral tilt to improve cardiac output and oxygen therapy to maintain an oxygen saturation of 94-98%.

### 6.2 Poisoning

In patients with carbon monoxide poisoning, measurement of blood carboxyhaemoglobin level is required. The half-life of carboxyhaemoglobin is decreased by high concentrations of oxygen so such patients should be treated with high oxygen concentrations. This can be achieved with high flow oxygen via a reservoir mask in spontaneously breathing patients and with endotracheal intubation and mechanical ventilation with 100% oxygen in comatose patients or those with severe impairment of consciousness.

### 6.3 Posture

Oxygenation is reduced in the supine position, so fully conscious patients should be nursed in the most upright posture possible (special consideration should be given to patients with spinal cord injury about the sustainability of upright positioning for respiratory function) unless:

- It causes significant discomfort that is distressing to the patient and cannot be managed (e.g. pain relief post-surgery)
- Immobilisation is required for suspected or actual skeletal or spinal trauma
- Patient is hypotensive
- Post seizure recover period

### 6.4 Domiciliary Oxygen

The Department of Health Western Australia (DoHWA) funds the provision of clinically appropriate domiciliary oxygen to eligible Western Australian who have chronic hypoxaemia.

The clinical indications and contraindications for the provision of DoHWA funded domiciliary oxygen therapy as set out in the DoHWA Domiciliary Oxygen Prescription Form.

**The indications for domiciliary oxygen therapy are,**

1. Respiratory conditions which are chronic where there is evidence of hypoxaemia.
2. Nocturnal oxygen
3. Ambulatory oxygen for profound exertional desaturation without resting hypoxia.
4. Palliative oxygen therapy
5. Maximally treated chronic heart failure with symptomatic central sleep apnoea in patient intolerant of a CPAP device

**The Contraindications for domiciliary oxygen therapy are.**

1. Current smokers or e cigarette users
2. Smoking not ceased within 6 weeks of prescription for short term oxygen therapy.
3. Patients without any evidence hypoxaemia at rest and /or exertion as defined by the indication criteria.
4. Patients who have not received adequate investigation or therapy relevant to their condition.

## 7. Patient monitoring and documentation

- Oxygen therapy must be documented in the health records (DMR)
- Individualised monitoring plan to be documented in the patients' health records as soon as is practicable. At a minimum the plan must consider:
    - Diagnosis
    - Presence of comorbidities and treatment

- Protocol requirements
- Any restriction to intervention associated with advanced health directives (AHD) or the like
- Hypoxaemia is a result of a disease process therefore improving the oxygenation saturation by itself will not change the disease state. Careful monitoring and further investigations may be indicated depending upon clinical circumstances.
- If signs of deterioration are noted and/or a patient is 'triggering' on observation and response charts, or if you or the patient, family member or carer are worried, initiate escalation of care process specific to your area and document the episode in the health records.
- Ensure early recognition of and response to hypoxaemia. If the patient meets medical emergency team calling criteria, initiate a medical emergency team (MET) call.

- Monitor oxygen therapy effect by:
    - Pulse oximetry or arterial blood gas measurements. (Pulse oximetry may not be accurate in patients with poor peripheral circulation)
    - Respiratory rate and distress
    - Level of consciousness
    - Capillary and venous (peripheral oxygen) $PvO_2$ measurements are not a substitute for arterial $PaO_2$.However, bicarbonate, peripheral carbon dioxide (PvCO2) and venous pH correlate with arterial measurements when collected correctly.

## 8. Oxygen and Oxygen Cylinders

Oxygen is supplied through an automated ward stock wall outlet system and is usually found as a white outlet on the back wall behind the patient's bed. An oxygen outlet flow meter is attached by way of delivering oxygen to the patient and controlling the flow rate. The oxygen is also delivered to patient during transfer by a portable integrated oxygen cylinder and flow regulator system called Inhalo™oxygen cylinder .The inhalo™ oxygen cylinder is a light weight ergonomic cylinder with an integrated valve, it has an in built regulator and flow meters and has a gauge to indicate the amount of gas left in the cylinder. The inhalo ™ cylinders are white in colour with a label on the neck to identify the contents

Inhalo™ oxygen cylinder comes in two sizes.

- 400 CD holding 630L.
- 400C holding 490L.


Oxygen Cylinder checking can be done ensuring the oxygen outlets are clean, oxygen is flowing and when the outlet meter is on the sound of the escaping oxygen can be heard and gas can be felt on the hand.

### 8.1 Checking of the inhalo™ oxygen cylinder

- Turn the open/close hand wheel greater than a quarter turn and then rotate the flow indicator knob to produce the flow of oxygen to confirm the cylinder is functioning correctly.
- Check that the pressure gauge is working properly by laying the cylinder down from a vertical position to a horizontal position, see the figure below.

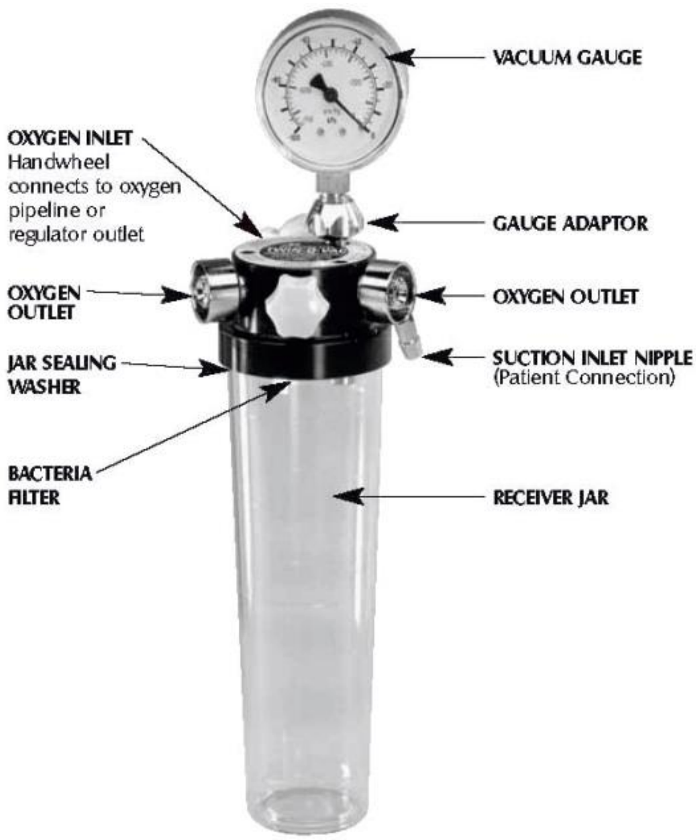
**Figure A**

✓

**Figure B**

✗

Figure A. The pointer reads the same cylinder pressure, the gauge is working as expected, indicating the gas contents of the cylinder. Please continue to use the 400CD cylinder. Figure B. The position of the pointer is still upwards, the cylinder pressure reading has changed, the pressure gauge is **FAULTY.**



If checking the cylinder on the resuscitation trolley, sign the attached checklist to indicate the check has been successfully completed. For the oxygen cylinder inhalo™ in resuscitation trolley a Twin-O-Vac system is also attached

## 9. Legislative context

The following documents are required to give effect to this policy:

- Department of Health Work Health and Safety Policy
- Department of Health Clinical Incident Management Policy MP 0122/19
- EMHS Occupational Safety and Health
- AKG Recognising and Responding to Acute Deterioration Policy
- HDWA Clinical Handover Policy MP 0095
- East Metropolitan Health Service (EMHS) Patient Identification and Procedure Matching Policy EMHS: 21

## 10. Supporting information

The following documents support the implementation of this policy:

- AKG Clinical Handover Procedure

## 11. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated by the appropriate designations via routine clinical incident review processes.

## 12. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1: Governance for Safety and Quality in Health Service Organisations
- Standard 2: Partnering with Consumers
- Standard 3: Preventing and Controlling Healthcare Associated Infections
- Standard 4: Medication Safety
- Standard 5: Comprehensive Care
- Standard 6: Communication for safety
- Standard 8: Recognising and responding to acute Deterioration.

## 13. References

1. Thoracic Society of Australia and New Zealand. (2016). TSANZ Domiciliary Oxygen Therapy Guidelines. Retrieved from TSANZ-DomiciliaryOxygen-Guidelines-2016-web.pdf (thoracic.org.au)
2. Barnett, A., Beasley, R., Buchan, C., Chien, J., Farah, C. S., King, G., McDonald, C. F., Miller, B., Munsif, M., Psirides, A., Reid, L., ... Thoracic Society of Australia and New Zealand. (2022). Thoracic Society of Australia and New Zealand position statement on acute oxygen use in adults: 'Swimming between the flags'. *Respirology*, *27*(8), 753–759. Thoracic Society of Australia and New Zealand Position Statement on Acute Oxygen Use in Adults: ‘Swimming between the flags’ - Barnett - 2022 - Respirology - Wiley Online Library

3. British Thoracic Society Emergency Oxygen Guideline Development Group. (2017). BTS guideline for oxygen use in adults in healthcare and emergency settings. *Thorax, 72*(Suppl. 1)
4. Hamilton Medical. (n.d.). e-Academy. Hamilton Medical. e-Academy | Hamilton Medical (hamilton-medical.com)
5. Fisher&Paykel Healthcare. (n.d.). Airvo™ 2 System. Retrieved from Airvo™ 2 Nasal High Flow/HFNC System [2 - 60 L/min] | Fisher & Paykel Healthcare (fphcare.com)
6. Government of Western Australia. (2019, November). CUAGAS2016 BOC Inhalo (400CD) Pre-use Check Flyer [PDF]. Retrieved from CUAGAS2016 BOC Inhalo (400CD) Pre-use Check Flyer_0.pdf (www.wa.gov.au)
7. Health Service Executive. (n.d.). Guidance for the Care of the Adult Patient with a Tracheostomy [PDF]. Retrieved from Guidance for the Care of the Adult Patient with a Tracheostomy (hse.ie)
8. Fiona Stanley Hospital Group, Government of Western Australia. (2022). Oxygen Therapy Management in Adults (Document No. FSH-HW-NPS-0076).
9. Royal Perth Bentley Group. (2023). Clinical Practice Standard for Oxygen Therapy Management (for Adult Inpatients).
10. Australian Commission on Safety and Quality in Health Care. (2019). National Safety and Quality Health Service Standards: Second Edition [PDF]. Retrieved from National Safety and Quality Health Service Standards, Second Edition
11. Air Liquide Healthcare. (2023). Twin-O-Vac Instructions for Use [Instruction Manual]. Retrieved from 521512-08-320.cdr (airliquide.com)


## 14. Document owner

**Clinical Nurse Specialist - Resuscitation**

Enquiries relating to this guideline may be directed to:


Title: CNS Resuscitation
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## 15. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

| Version | Effective from | Effective to | Amendment(s) |
|---|---|---|---|
| 3 | 26/07/2024 | 26/07/2027 | |

## 16. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

| | |
|---|---|
| **Authorised by** | Tania Strickland A/Director Clinical Services |
| **Authorisation date** | 26/07/2024 |
| **Published date** | 26/07/2024 |